

Hossein Hatamnia

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Education

University of California, Riverside (UCR)

Ph.D. in Astronomy

• Advisors: Bahram Mobasher & Drew Newman

Riverside, California, US

Sep 2024 - Present (Expected 2030)

University of California, Riverside (UCR)

M.S. in Astronomy (en route)

• GPA: 3.98/4

Riverside, California, US

Mar 2026

Sharif University of Technology (SUT)

B.Sc. in Physics

• GPA: 18.56/20

Tehran, Iran

Sept 2019-Feb 2024

Research Interests

I study how the cosmic web environment drives galaxy formation and evolution, tracing the interplay between large-scale structure and galaxy properties out to high redshift. I also work on applying AI methods to astronomical data analysis and classification.

Achievements & Fellowships

- 2026** UCR-Carnegie Observatory 3-Year **Fellowship** (highly competitive graduate fellowship supporting Ph.D. research in collaboration with Carnegie Observatories. Only one student each year.)
- 2024** Dean's **Fellowship**, University of California, Riverside (graduate recruitment fellowship)
- 2026** The Junior Graduate Award (departmental award recognizing outstanding graduate achievement)
- 2024** Institute for Research in Fundamental Sciences (IPM) **Fellowship**
- 2018** **Silver Medal**, Iran's National Astronomy and Astrophysics Olympiad
- 2019** **Top 1% Nationwide**, Iranian National University Entrance Examination

Publications (12 Papers & 217 Citations)

- 1. Hatamnia, H.,** Mobasher, B., Taamoli, S., Kartaltepe, J. S., Casey, C. M., et al. (2026). Large-scale Structure in COSMOS-Web: Tracing Galaxy Evolution in the Cosmic Web up to $z \sim 7$ with the Largest JWST Survey. *The Astrophysical Journal*, **1002**(2), Article 192. [doi:10.3847/1538-4357/ae5bac](https://doi.org/10.3847/1538-4357/ae5bac) **Citations: 3**
- 2. Scognamiglio, D.,** Leroy, G., Harvey, D., Massey, R., Rhodes, J., **Hatamnia, H.,** et al. (2026). An ultra-high-resolution map of (dark) matter. *Nature Astronomy*. [doi:10.1038/s41550-025-02763-9](https://doi.org/10.1038/s41550-025-02763-9) **Citations: 3**
- 3. Tanaka, T. S.,** Akins, H. B., Harikane, Y., Silverman, J. D., Casey, C. M., **Hatamnia, H.,** et al. (2025). Discovery of a Little Red Dot Candidate at $z \gtrsim 10$ in COSMOS-Web Based on MIRI-NIRCam Selection. *The Astrophysical Journal*, 995(1), Article 21. [doi:10.3847/1538-4357/ae145f](https://doi.org/10.3847/1538-4357/ae145f) **Citations: 24**
- 4. Shuntov, M.,** Akins, H. B., Paquereau, L., Casey, C. M., Ilbert, O., **Hatamnia, H.,** et al. (2025). COSMOS2025: The COSMOS-Web galaxy catalog of photometry, morphology, redshifts, and physical parameters from JWST, HST, and ground-based imaging. *Astronomy & Astrophysics*, 704, A339. [doi:10.1051/0004-6361/202555799](https://doi.org/10.1051/0004-6361/202555799) **Citations: 81**
- 5. Khostovan, A. A.,** Kartaltepe, J. S., Salvato, M., Ilbert, O., Casey, C. M., **Hatamnia, H.,** et al. (2026). COSMOS Spectroscopic Redshift Compilation (First Data Release): 488,000 Redshifts Encompassing Two Decades of Spectroscopy. *The Astrophysical Journal Supplement Series*, 282(1), 6. [doi:10.3847/1538-4365/ae1cb9](https://doi.org/10.3847/1538-4365/ae1cb9) **Citations: 65**

6. Khoshtinat, M., **Hatamnia, H.**, & Baghram, S. (2024).

One-point statistics in various cosmic environments in the presence of massive neutrinos.

Monthly Notices of the Royal Astronomical Society, 534(2), 1166–1174. doi:10.1093/mnras/stae2140

Citations: 2

Media Coverage

Hatamnia et al. 2026 — Large-scale Structure in COSMOS-Web (First Author)

UCR News, RIT news, Space.com, Live Science, ScienceDaily, MSN, Phys.org, (+10 more outlets, May 2026)

Scognamiglio et al. 2026 — Ultra-High-Resolution Dark Matter Map, *Nature Astronomy*

NASA.gov, National Geographic, CBS News, Scientific American, Smithsonian Magazine, BBC (+10 more outlets, Jan 2026)

Shuntov et al. 2025 — COSMOS2025 Galaxy Catalog

Space.com, Discover Magazine, Phys.org, RIT News, UC Santa Barbara News (+10 more outlets, Jun 2025)

Selected Talks & Poster Presentations

Large-Scale Structure in COSMOS-Web: Evolution of Galaxies in the Cosmic Web

Marseille, France

COSMOS Meeting

May 2025

- Contributed Talk

Radio Spectral Energy Distribution of Star Forming Galaxies at $z < 1.5$

Bristol University, Bristol, UK

Towards SKAO (MeerKAT Symposium and Workshop)

Sept 2023

- Poster Presentation

Radio Spectral Energy Distribution of Star Forming Galaxies at $z < 1.5$

Erciyes University, Kayseri, Turkey

IAU/IHOW Radio Astronomy Workshop

Sept 2023

- Contributed Talk

Research & Teaching Experiences

Teaching Assistant

University of California, Riverside

Undergraduate Physics Laboratories

Since Winter 2025

Research Assistant

Institute of Research in

Fundamental Sciences (IPM)

Supervisor: Dr. Fatemeh Tabatabaei

Apr 2024 - Sept 2024

As a member of the MIGHTEE team (PI: Matt Jarvis, Oxford), I use Bayesian MCMC to model radio spectral energy distribution.

Teaching Assistant

Sharif University of Technology

Astrophysics I

Spring 2023

Teaching Assistant

Sharif University of Technology

Analytical Mechanics I, II

Sept 2022 - Jul 2023

Teaching Assistant

Sharif University of Technology

Basic programming with Python

Spring 2021

Teacher

High Schools

Astronomy Olympiad

Feb 2020 - Apr 2024

Selected Courses

Data Analysis & High-Performance Computing **Grade: 18.9/20**

Astrophysics I, II **Grade: 18.5, 19/20**

Computational Astrophysics **Grade: 4/4**

Observational Astronomy **Grade: 4/4**

Cosmology **Grade: 4/4**

Gravity and General Relativity **Grade: 18.4/20**

Nonlinear Dynamics and Chaos **Grade: 20/20**

Stellar Structure **Grade: 4/4**

Skills

Programming

Python, Julia, C/C++

Scientific Libraries

NumPy, SciPy, Astropy, Matplotlib, OpenCV, LinearAlgebra, emcee

Deep Learning

TensorFlow, PyTorch, Keras, Flux, MLBase

Astronomy Software

EAZY, Bagpipes, DS9, CASA, CARTA, AIPS

High-Performance Computing

Multi-Core Computing, Multi-Threading, CUDA

Dev Tools

Linux, Git, LaTeX